

BTYT Banking Analytics --- Mathematical and Statistical Foundations

Project: Banco de Treinta y Tres (BTYT) Banking Analytics Document type: Living mathematical specification and study reference Status: Active --- updated as generators are refactored and frozen Current phase: Part I architecture refactor, before canonical large-scale generation Last updated: 2026-09-03

1. Purpose and scope

This document records the mathematical, probabilistic, statistical, temporal, and accounting foundations of BTYT.

It is a living mathematical specification. Its purpose is not to decorate the project with mathematical notation, but to make explicit the mechanisms that actually govern the synthetic banking universe.

Three levels of formalization are distinguished throughout the document:

- Implemented: confirmed by production BTYT code.
- Conceptual: a mathematical representation of an implemented

design principle, even when the Python code expresses it procedurally rather than through the exact displayed equation.

- Planned: mathematics expected in a later phase but not yet implemented.

A convenient textbook model must never be presented as an implemented BTYT mechanism unless the code actually uses it.

The central modeling principle is:

BTYT models causes as probabilistic shifts in behavior, not deterministic assignments of outcomes. Shared causal conditions may influence multiple processes, but downstream realizations use independent stochastic streams.

PART A --- GLOBAL MATHEMATICAL NOTATION

2. Index notation

The following indices are preferred throughout BTYT.

Symbol	Meaning
i	Customer or generic individual entity
a	Account
b	Bank
r	Branch
l	Loan
c	Campaign or category, depending on context
k	Product, segment, category, or generic
t	discrete state
time	period
m	Month
y	Year
n	Monte Carlo world / simulation index
j	RNG stream or generic
secondary index	
N	Population or sample size

When a symbol could be ambiguous, the corresponding section defines it explicitly.

3. Preferred Greek notation

Symbol Preferred BTYT interpretation ----- (π) Probability associated with a modeled event

(μ) Mean, baseline, or long-run level

(σ) Standard deviation or volatility

(σ²) Variance

(φ) Temporal persistence / autoregressive coefficient

(λ) Event intensity or rate

(α) Intercept, baseline effect, or entity effect

(β) Primary sensitivity / coefficient

(γ) Secondary sensitivity / coefficient

(δ) Additional structural effect

(θ) Shock or interaction sensitivity

(ρ) Correlation or dependence parameter

(ε) Idiosyncratic innovation / random disturbance

(η) Additional latent disturbance or entity-specific shock

(ω) Weight

(τ) Time threshold or event time

(κ) Threshold, scaling, or concentration parameter when required

These conventions reduce arbitrary notation and make the mathematical specification easier to extend.

PART B --- PROBABILITY AND STOCHASTIC GENERATION

4. Random variables and realizations

Let (X) denote a random variable and (x) one realized value.

A generic stochastic mechanism is written:

$$X \sim F(\boldsymbol{\theta})$$

where:

- F is a probability distribution;
- $\boldsymbol{\theta}$ is its parameter vector.

A BTYT generator therefore defines a mechanism that produces a realization of the world rather than merely assigning a fixed collection of values.

5. Bernoulli events

For a binary event associated with entity (i):

$$Y_i \mid \pi_i \sim \text{Bernoulli}(\pi_i)$$

with:

$$P(Y_i = 1) = \pi_i$$

and:

$$P(Y_i = 0) = 1 - \pi_i$$

where:

$$0 \leq \pi_i \leq 1$$

This general structure is relevant to stochastic events such as closures, failures, campaign responses, operational degradation, and other binary realizations.

6. Conditional probability and heterogeneity

BTYT generally avoids assigning the same probability to every entity.

Instead:

$$\pi^*_{i,t} = P(Y^*_{i,t} = 1 \mid \mathbf{x}_{i,t}, \mathbf{z}_t)$$

where:

- $(\mathbf{x}_{i,t})$ contains entity-specific information;
- $(\mathbf{z}_t)$ contains shared conditions at time (t).

Thus:

$$\pi^*_{i,t} \neq \pi^*_{j,t}$$

may hold for two different entities even during the same period.

Likewise:

$$\pi^*_{i,t} \neq \pi^*_{i,t+1}$$

may hold for the same entity across time.

7. Probabilistic causal shifts

Suppose a baseline event probability is (π_0) .

A conceptual additive latent specification can be represented as:

$$\pi^*_{i,t} = \pi^*_0 + \alpha^*_{i,t} + \beta^*_{k(i)} + \gamma^*_{t} + \delta^*_{g(i)} + \theta S_t + \varepsilon^*_{i,t}$$

where:

- $(\alpha_{i,t})$ = individual effect;
- $(\beta_{k(i)})$ = product or segment effect;
- (γ_t) = temporal effect;
- $(\delta_{g(i)})$ = geographic effect;
- (S_t) = systemic shock;
- (θ) = sensitivity to the shock;
- $(\varepsilon_{i,t})$ = idiosyncratic component.

If this representation is used directly as a probability, it must be bounded:

$$\pi^*_{i,t} = \min(1, \max(0, \pi^*_{i,t^*}))$$

This equation is conceptual unless confirmed by a specific generator. Some BTYT generators use multiplicative adjustments, normalized weights, or other transformations instead.

The invariant is the causal principle:

$$S_t \Rightarrow \Delta \pi_{i,t}$$

but not:

$$S_t \Rightarrow Y_{i,t} = 1$$

8. Weighted categorical distributions

For a categorical variable (X_i) taking one of (K) states:

$$X_i \sim \text{Categorical}(\omega_1, \omega_2, \dots, \omega_K)$$

subject to:

$$\omega_k \geq 0$$

and:

$$\sum_{k=1}^K \omega_k = 1$$

This is the generic mathematical structure behind weighted selections among discrete alternatives.

PART C --- RNG AND REPRODUCIBILITY

9. Seeded pseudorandom generation

Let (s) denote the canonical seed.

A pseudorandom sequence can be represented as:

$$\mathcal{R} = G(s)$$

where (G) is the pseudorandom generator.

For the same implementation:

$$s_1 = s_2 \implies \mathcal{R}_1 = \mathcal{R}_2$$

provided that the draw order and algorithm remain unchanged.

10. Independent stochastic streams

For stream (j):

$$\mathcal{R}_j = G(s, j)$$

The purpose of (j) is to separate conceptually different stochastic processes.

Ideally:

$$\mathcal{R}_j \perp \mathcal{R}_k \quad j \neq k$$

in the practical design sense that downstream processes consume separate deterministic streams derived from the common world seed.

This supports the BTYT principle:

shared causes + independent realizations

11. Chunk-invariance requirement

Let:

$$\mathcal{W}(s, \mathbf{c}, h)$$

denote the realized world generated with:

- seed (s);
- world configuration (c);
- technical chunk size (h).

The desired invariant is:

$$\mathcal{W}(s, \mathbf{c}, h_1) = \mathcal{W}(s, \mathbf{c}, h_2)$$

for valid chunk sizes (h_1) and (h_2).

Therefore:

Chunk size is an engineering parameter, not a statistical parameter.

The exact implementation required to guarantee this invariant will be established during generator refactoring.

PART D --- TEMPORAL DYNAMICS

12. Longitudinal variables

A time-varying entity-level variable is written:

$$X_{i,t}$$

while a system-wide variable is written:

$$X_t$$

Depending on the subsystem, (t) may represent a date, month, or year.

13. Autoregressive persistence

A standard first-order persistent process is:

$$X_t - \mu = \phi(X_{t-1} - \mu) + \varepsilon_t$$

or equivalently:

$$X_t = (1 - \phi)\mu + \phi X_{t-1} + \varepsilon_t$$

where:

- (μ) = long-run level;
- (ϕ) = persistence;
- (ε_t) = innovation.

Under the conventional stationary AR(1) case:

$$|\phi| < 1$$

This notation is useful for BTYT's AR-like market and macro dynamics. Exact equations will replace conceptual forms as generators are frozen.

14. Innovations and volatility

A common innovation assumption can be written:

$$\varepsilon_t \sim F(0, \sigma^2)$$

and, when the implementation specifically uses a Gaussian innovation:

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

The Gaussian form must not be assumed unless confirmed by code.

15. Shared shocks

Let (S_t) be a systemic shock.

An entity-level state can be represented conceptually as:

$$X_{i,t} = \mu_{i,t} + \beta^* i S_t + \varepsilon^* i, t$$

where:

- (β_i) = exposure/sensitivity;

- $(\varepsilon_{\{i,t\}})$ = idiosyncratic innovation.

Two entities can therefore experience the same shock differently:

$$\beta_i \neq \beta_j$$

16. Interaction effects

If the effect of (X_1) depends on (X_2) , a generic interaction term is:

$$\beta_3 X_1 X_2$$

within a conceptual model such as:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

This notation represents the interaction concept and does not imply that BTYT necessarily estimates a linear regression for the corresponding generator.

PART E --- BANKING MARKET MATHEMATICS

17. Market-share simplex

Let:

$$\omega_{b,t}$$

be the market share of bank (b) at time (t).

Then:

$$\omega_{b,t} \geq 0$$

and:

$$\sum_{b=1}^B \omega^*_b = 1, t = 1$$

The vector:

$$\boldsymbol{\omega}^*_t = (\omega^*_1, \omega^*_2, \dots, \omega^*_B)$$

therefore belongs to the probability simplex:

$$\boldsymbol{\omega}_t \in \Delta^{B-1}$$

This normalization constraint is structural.

18. Bank-specific boundaries

Let:

$$\ell_b$$

denote the lower plausible boundary for bank (b), and:

$$u_b$$

the upper boundary.

Then:

$$\ell_b \leq \omega_b \leq u_b$$

Available headroom is:

$$h_b = u_b - \ell_b$$

A normalized headroom preference is:

$$q_b = \frac{h_b}{\sum_{j=1}^B h_j}$$

with:

$$\sum_b q_b = 1$$

This notation corresponds to the headroom normalization concept used during bank-market calibration.

19. Dynamic latent bank position

A useful conceptual representation of bank (b)'s latent market position is:

$$z_{b,t} = \mu_b + \phi * b(z^* b, t - 1 - \mu_b) + \beta_b M_t + \gamma * b S_t + \varepsilon * b, t$$

where:

- (μ_b) = bank-specific long-run position;
- (ϕ_b) = persistence;
- (M_t) = common market/macro state;
- (β_b) = sensitivity to the common state;
- (S_t) = shock;
- (γ_b) = shock sensitivity;
- ($\varepsilon_{\{b,t\}}$) = bank-specific innovation.

The realized market weights must then satisfy the simplex and boundary constraints.

The exact production transformation from latent state to ($\omega_{\{b,t\}}$) will be documented from the final bank generator.

PART F --- MONTE CARLO AUDITING

20. Simulated worlds

Let:

$$\mathcal{W}^{(1)}, \mathcal{W}^{(2)}, \dots, \mathcal{W}^{(N)}$$

be (N) independently seeded or stream-controlled simulated worlds used for auditing.

For statistic (X):

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N X^{(n)}$$

21. Empirical event probability

Let $\mathbb{1}(A)$ denote the indicator function:

$$\mathbb{1}(A) = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{if } A \text{ does not occur} \end{cases}$$

Then:

$$\hat{P}(A) = \frac{1}{N} \sum_{n=1}^N \mathbb{1}(A_n)$$

Examples include:

- a bank finishing first;
- a bank reaching the top three;
- a boundary being hit;
- a branch closing;
- a guardrail failing.

22. Empirical quantiles

For simulated statistic (X), BTYT frequently uses quantiles such as:

$$Q_{0.05}(X), \quad Q_{0.50}(X), \quad Q_{0.95}(X)$$

corresponding to P05, median, and P95.

These describe the range of plausible worlds rather than only the canonical realization.

PART G --- BRANCH MATHEMATICS

23. Stochastic closures

For branch (r):

$$C_r \mid \pi_r^{(C)} \sim \text{Bernoulli}(\pi_r^{(C)})$$

where:

$$C_r = \begin{cases} 1, & \text{branch closes} \\ 0, & \text{branch remains open} \end{cases}$$

and:

$$\pi_r^{(C)} = f(\mathbf{x}_r, \mathbf{z}_t)$$

depends on branch and world conditions.

A shock may produce:

$$\pi^{*r, \text{shock}}^{(C)} \pi^{*r, \text{baseline}}^{(C)}$$

without forcing:

$$C_r = 1$$

24. Number of closures per world

For simulated world (n):

$$K^{(n)} = \sum_{r=1}^R C_r^{(n)}$$

Monte Carlo auditing examines the empirical distribution:

$$\hat{P}(K = k)$$

as well as branch-specific closure frequencies:

$$\hat{\pi}^{*r(C)} = \frac{1}{N} \sum_{n=1}^N C_r^{(n)}$$

PART H --- CUSTOMERS AND PRODUCT RELATIONSHIPS

25. Hierarchical generation

The synthetic banking universe contains conditional relationships rather than independent flat tables.

Conceptually:

$$Customer_i \rightarrow Account_{i,a} \rightarrow Card_{i,a,k}$$

and:

$$(Customer_i, Account_{i,a}, Product_k) \rightarrow Loan_{i,l}$$

Thus:

$$P(A_{i,a} \mid X_i)$$

and:

$$P(L_{i,l} \mid X_i, A_{i,a}, P_k)$$

can depend on upstream state.

This hierarchical structure creates economically meaningful heterogeneity for later analytics.

PART I --- CREDIT MATHEMATICS

26. Loan stock-flow identity

Let:

- $(B_{\{l,t\}})$ = loan balance;
- $(I_{\{l,t\}})$ = interest accrued;
- $(P_{\{l,t\}})$ = principal/payment reduction;
- $(A_{\{l,t\}})$ = other explicitly modeled adjustments.

A generic balance evolution is:

$$B_{l,t} = B_{l,t-1} + I_{l,t} - P_{l,t} + A_{l,t}$$

The exact BTYT product-specific implementation will be documented when the credit generators are frozen.

27. Payment shortfall

Let:

$$S_{l,t}$$

be scheduled payment and:

$$A_{l,t}^{(pay)}$$

actual payment.

Then:

$$H_{l,t} = \max(0, S_{l,t} - A_{l,t}^{(pay)})$$

where $(H_{l,t})$ is payment shortfall.

28. Delinquency state

Let:

$$d_{l,t} \geq 0$$

be days past due.

A delinquency state can be represented as:

$$D_{l,t} = g(d_{l,t})$$

where (g) maps days past due into risk buckets.

The exact BTYT thresholds will be inserted from the final lifecycle implementation.

29. Stochastic default

For loan (l):

$$Y_l^{(D)} \mid \pi_l^{(D)} \sim \text{Bernoulli}(\pi_l^{(D)})$$

with:

$$\pi_l^{(D)} = f(\mathbf{x}_l, \mathbf{p} * l, \mathbf{h} * l, t, \mathbf{m} * t, \mathbf{s} * l, t)$$

where the inputs may represent customer characteristics, product/origination information, historical behavior, macro conditions, and shocks.

The essential requirement is:

$$0 < \pi_l^{(D)} < 1$$

for non-degenerate cases.

Risk characteristics should alter default probability rather than perfectly determine default.

30. Information set and leakage

For a prediction at time (t), define:

$$\mathcal{I}_t$$

as the information available at that time.

A valid feature must satisfy:

$$X \in \mathcal{I}_t$$

Future information:

$$X_{t+h}, \quad h \geq 0$$

must not enter the feature set if it reveals information unavailable at prediction time.

This is the formal basis for avoiding target leakage in Part II.

PART J --- TRANSACTION AND BALANCE MATHEMATICS

31. Signed transaction flow

For account (a) and period (t):

$$F_{a,t} = I_{a,t} - O_{a,t}$$

where:

- $(I_{a,t})$ = completed inflows;
- $(O_{a,t})$ = completed outflows.

32. Balance reconciliation

Let:

- $(B_{a,t}^{(0)})$ = starting balance;
- $(B_{a,t}^{(1)})$ = ending balance.

Then:

$$B_{a,t}^{(1)} = B_{a,t}^{(0)} + I_{a,t} - O_{a,t} + A_{a,t}$$

where $(A_{a,t})$ contains only explicitly modeled adjustments outside the ordinary transaction-flow terms.

When no such adjustment exists:

$$B_{a,t}^{(1)} = B_{a,t}^{(0)} + I_{a,t} - O_{a,t}$$

This is an accounting identity, not a statistical expectation.

33. Balance continuity

For consecutive periods:

$$B_{a,t+1}^{(0)} = B_{a,t}^{(1)}$$

subject only to explicitly modeled account lifecycle boundaries.

34. Internal transfers

For transfer amount ($q > 0$) from account (a) to account (b):

$$\Delta B_a = -q$$

and:

$$\Delta B_b = +q$$

Therefore:

$$\Delta B_a + \Delta B_b = 0$$

for the principal transfer amount.

This is the conservation identity underlying internal-transfer reconciliation.

35. Failed transactions

For a failed transaction with nominal amount (q), the transaction itself should not generate the completed-flow balance effect:

$$Status = FAILED \implies \Delta B^{(q)} = 0$$

subject to any separately modeled fees or operational effects.

36. Transaction-count intensity

A generic count-process notation is:

$$N_{a,t} \sim \mathcal{C}(\lambda_{a,t}, \boldsymbol{\theta})$$

where:

- $\mathcal{C}$ = the actual count mechanism;
- $\lambda_{a,t}$ = activity intensity;
- $\boldsymbol{\theta}$ = any additional parameters.

Conceptually:

$$\lambda_{a,t} = \lambda_a \cdot s_t \cdot m_{a,t}$$

where:

- λ_a = baseline account intensity;
- s_t = seasonal effect;
- $m_{a,t}$ = behavioral/macro multiplier.

The notation deliberately avoids claiming a Poisson distribution until the final transaction generator confirms the exact mechanism.

37. Channel evolution

For digital channel usage:

$$\pi_{i,t}^{(digital)} = P(\text{Channel} * i, t = \text{Digital})$$

and for cash usage:

$$\pi_{i,t}^{(cash)} = P(\text{Channel} * i, t = \text{Cash})$$

These may evolve as:

$$\pi_{i,t}^{(digital)} = f(t, \mathbf{x}_i, \mathbf{z}_t)$$

and:

$$\pi_{i,t}^{(cash)} = g(t, \mathbf{x}_i, \mathbf{z}_t)$$

allowing heterogeneous digital adoption and cash transition over time.

PART K --- EXTERNAL SHOCK MATHEMATICS

38. Systemic and idiosyncratic shock decomposition

Let:

$$S_t$$

be the systemic shock and:

$$\eta_{i,t}$$

an idiosyncratic shock.

A combined shock state can be represented as:

$$Z_{i,t} = S_t + \eta_{i,t}$$

The exact production aggregation may differ and will be documented from the final generator.

39. Exposure and resilience

Let:

- $(e_{\{i,t\}})$ = exposure;
- $(r_{\{i,t\}})$ = resilience;
- $(Z_{\{i,t\}})$ = shock state.

A conceptual impact function is:

$$\Psi^* i, t = Z^* i, t \cdot e_{i,t} \cdot (1 - r_{i,t})$$

with larger resilience reducing effective impact.

This is conceptual notation unless the frozen generator uses this exact transformation.

40. Cross-system causal structure

A common shock may influence several probabilities:

$$\pi^* i, t^{(D)} = f_D(\mathbf{x}^* i, t, S_t)$$

$$\pi^* i, t^{(A)} = f_A(\mathbf{x}^* i, t, S_t)$$

$$\pi^* r, t^{(C)} = f_C(\mathbf{x}^* r, t, S_t)$$

where the superscripts can represent default, activity, and closure.

However:

$$Y_{i,t}^{(D)}, \quad Y_{i,t}^{(A)}, \quad Y_{r,t}^{(C)}$$

remain separate stochastic realizations.

Therefore common causation does not imply identical outcomes.

PART L --- CAMPAIGN MATHEMATICS

41. Eligibility, exposure, and response

For customer (i) and campaign (c), define:

- $(E_{\{i,c\}})$ = eligibility;
- $(X_{\{i,c\}})$ = exposure;

- $(R_{i,c}) = \text{response}$.

The logical sequence is:

$$E_{i,c} \rightarrow X_{i,c} \rightarrow R_{i,c}$$

For attributed response times:

$$\tau_{i,c}^{(R)} \geq \tau_{i,c}^{(X)}$$

42. Response probability

Conditional on exposure:

$$R_{i,c} \mid X_{i,c} = 1 \sim \text{Bernoulli}(\pi_{i,c}^{(R)})$$

where:

$$\pi_{i,c}^{(R)} = f(\mathbf{x}_i, \mathbf{c}, \text{channel}, \text{geography}, \text{time}, \text{behavior}, \text{externalstate})$$

Targeting affects response probability; it does not deterministically assign response.

43. Response rate

If:

$$N_c^{(X)} = \sum_i X_{i,c}$$

and:

$$N_c^{(R)} = \sum_i R_{i,c}$$

then:

$$RR_c = \frac{N_c^{(R)}}{N_c^{(X)}}$$

for $(N_c^{(X)} > 0)$.

Conditional response rates can be calculated analogously by channel, geography, segment, or other dimensions.

PART M --- OPERATIONAL DATA RELIABILITY

44. Degradation probability

For record (i):

$$Q_i \mid \pi_i^{(Q)} \sim \text{Bernoulli}(\pi_i^{(Q)})$$

where:

$$\pi_i^{(Q)} = f(\text{incident}, \text{system}, \text{time}, \text{reliabilitylevel}, \text{recordcharacteristics})$$

Thus:

$$\text{Incident} = \mathcal{X} \Rightarrow Q_i = 1$$

Instead:

$$\text{Incident} = 1 \Rightarrow \Delta \pi_i^{(Q)}$$

45. Truth and operational observation

Let:

$$X_i^*$$

be synthetic truth and:

$$\widetilde{X}_i$$

the operationally observed value.

For protected variables:

$$\widetilde{X}_i = X_i^*$$

For degradable metadata:

$$\widetilde{X}_i = \mathcal{D}(X_i^*, Q_i)$$

where ($\mathcal{D}$) is the degradation operator.

This formal distinction allows BTYT to know the true synthetic state while also modeling imperfect operational data.

PART N --- VALIDATION MATHEMATICS

46. Referential integrity

Let ($\mathcal{P}$) be the set of valid parent keys.

For every required foreign key (f_i):

$$f_i \in \mathcal{P}$$

The invalid-key count is:

$$N_{FK}^{invalid} = \sum_{i=1}^N \mathbb{1}(f_i \notin \mathcal{P})$$

A strict PASS requires:

$$N_{FK}^{invalid} = 0$$

47. Primary-key uniqueness

For primary key (k_i):

$$\|k_1, \dots, k_N\| = N$$

or equivalently:

$$N_{PK}^{duplicate} = 0$$

48. Temporal integrity

For causally ordered events:

$$\tau_1 \leq \tau_2$$

Examples include:

$$\tau_{origination} \leq \tau_{payment}$$

and:

$$\tau_{exposure} \leq \tau_{response}$$

Lifecycle events must also fall inside the valid existence interval of the relevant entity.

49. Reconciliation residuals

Let:

- (O_i) = observed value;
- (E_i) = value implied by an accounting or structural identity.

Define residual:

$$e_i = O_i - E_i$$

Absolute residual:

$$\|e_i\|$$

Maximum absolute residual:

$$e_{\max} = \max_i \|e_i\|$$

For exact identities, the target is:

$$e_{\max} = 0$$

apart from explicitly accepted floating-point tolerance.

50. Descriptive moments

For observations $(x_1, \dots, x_N)$:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Sample variance:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

Sample standard deviation:

$$s = \sqrt{s^2}$$

These statistics complement quantiles and empirical frequencies in generator auditing.

PART O --- SCALE AND PERFORMANCE MATHEMATICS

51. Throughput

For (R) generated rows over elapsed time (T):

$$\nu = \frac{R}{T}$$

where (ν) is throughput, typically measured in rows per second.

52. Population and runtime scaling

For two benchmark runs:

$$\chi_N = \frac{N_2}{N_1}$$

and:

$$\chi_T = \frac{T_2}{T_1}$$

If:

$$\chi_T \approx \chi_N$$

runtime is approximately linear with customer population over the observed interval.

This must be measured empirically.

53. Empirical computational scaling

A useful benchmark relationship is:

$$T(N) \approx \alpha N^\beta$$

Taking logarithms:

$$\ln T = \ln \alpha + \beta \ln N$$

Interpretation:

$$\beta \approx 1$$

suggests approximately linear scaling;

$$\beta > 1$$

suggests superlinear scaling;

$$\beta < 1$$

suggests sublinear scaling over the measured range.

The parameter (β) here describes computational scaling and is unrelated to behavioral coefficients used elsewhere.

54. Storage projection

For observed size (S_1) at scale (N_1), a first-order projection is:

$$\hat{S}_2 = S_1 \frac{N_2}{N_1}$$

when storage scales approximately linearly with the relevant population.

For transaction tables, a better projection may use transaction row counts:

$$\hat{S} * tx, 2 = S * tx, 1 \frac{R_{tx, 2}}{R_{tx, 1}}$$

rather than customer population alone.

55. Compression

For CSV size ($S_{\{CSV\}}$) and Parquet size ($S_{\{PQ\}}$):

$$\kappa_{comp} = \frac{S_{CSV}}{S_{PQ}}$$

is the compression ratio.

Relative storage reduction is:

$$\delta_{storage} = 1 - \frac{S_{PQ}}{S_{CSV}}$$

These quantities will be measured during the scale ladder rather than assumed.

PART P --- PART II: CREDIT-SCORING MATHEMATICS

56. Status

The following sections are planned mathematical foundations for Part II. They should be expanded and marked as implemented only when the corresponding models are actually developed.

57. Binary target

Let:

$$Y_i = \begin{cases} 1, & \text{default} \\ 0, & \text{non-default} \end{cases}$$

The modeling objective is to estimate:

$$\pi_i^{(D)} = P(Y_i = 1 \mid \mathbf{x}_i)$$

58. Logistic regression

If logistic regression is retained:

$$\pi_i = \frac{1}{1 + \exp(-z_i)}$$

with:

$$z_i = \beta_0 + \sum_{j=1}^K \beta_j x_{ij}$$

The odds are:

$$\mathcal{O}_i = \frac{\pi_i}{1 - \pi_i}$$

and:

$$\ln \mathcal{O}_i = \beta_0 + \sum_{j=1}^K \beta_j x_{ij}$$

Thus:

$$e^{\beta_j}$$

is the multiplicative change in the odds associated with a one-unit increase in (x_j) , holding the remaining regressors constant.

59. Classification threshold

Given predicted probability $(\hat{\pi}_i)$ and threshold (τ) :

$$\hat{Y}_i = \mathbb{1}(\hat{\pi}_i \geq \tau)$$

The threshold is a decision parameter and need not equal (0.5).

In credit risk, threshold selection should reflect the relative costs of false positives and false negatives.

60. Confusion-matrix metrics

Let:

- (TP) = true positives;
- (TN) = true negatives;
- (FP) = false positives;
- (FN) = false negatives.

Then:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$F_1 = 2 \frac{Precision \cdot Recall}{Precision + Recall}$$

Accuracy, if used, is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

but it should not be interpreted in isolation, especially under class imbalance.

61. ROC and AUC

For threshold (τ):

$$TPR(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)}$$

and:

$$FPR(\tau) = \frac{FP(\tau)}{FP(\tau) + TN(\tau)}$$

The ROC curve plots:

$TPR(\tau)$ against $FPR(\tau)$

across thresholds.

ROC-AUC can be interpreted as the probability that a randomly selected positive observation receives a higher score than a randomly selected negative observation under the standard continuous-score interpretation.

62. Calibration

A probabilistic model is calibrated when:

$$P(Y = 1 \mid \hat{\pi} = p) \approx p$$

In practice, this is evaluated over probability ranges rather than requiring exact repeated probabilities.

For credit risk, calibration matters because the predicted probability itself has economic interpretation.

63. Brier score

For predicted probability ($\hat{\pi}_i$):

$$BS = \frac{1}{N} \sum_{i=1}^N (\hat{\pi}_i - Y_i)^2$$

Lower values indicate better probabilistic accuracy.

64. Expected credit loss bridge

If BTYT later extends risk analysis beyond probability of default, a natural banking decomposition is:

$$ECL = PD \times LGD \times EAD$$

where:

- (PD) = probability of default;
- (LGD) = loss given default;
- (EAD) = exposure at default.

This is not currently part of the committed Part II implementation. It is recorded only as a possible future mathematical extension if the project scope justifies it.

PART Q --- MATHEMATICAL GOVERNANCE

65. Exact versus conceptual equations

Every equation in this document should eventually be tagged mentally as one of:

Exact

The production generator implements the stated mechanism.

Equivalent

The implementation is procedural, but the equation is mathematically equivalent.

Conceptual

The equation explains the design but is not the exact production algorithm.

Planned

The mechanism belongs to a future phase.

As the refactor progresses, conceptual sections should be replaced by exact or equivalent specifications wherever possible.

66. Generator freeze protocol

When a generator is frozen:

1. inspect the final production code;
2. identify every random variable;
3. identify its actual distribution or draw mechanism;
4. identify deterministic transformations;
5. identify temporal recursions;
6. identify normalization and boundaries;
7. identify dependencies and interactions;
8. identify RNG streams;
9. identify hard accounting/relational constraints;
10. identify audit statistics;
11. update this document.

This creates a direct relationship:

Python implementation $\leftrightarrow$ mathematical specification

67. Three mathematical layers of BTYT

BTYT combines three distinct forms of mathematical structure.

Layer 1 --- Hard identities

These must hold.

Examples:

$$B_{a,t}^{(1)} = B_{a,t}^{(0)} + I_{a,t} - O_{a,t}$$

and:

$$\sum^* b \omega^* b, t = 1$$

Layer 2 --- Stochastic structure

These define probabilities and distributions.

Examples:

$$Y_{i,t} \sim \text{Bernoulli}(\pi_{i,t})$$

and:

$$\pi^* i, t = f(\mathbf{x}^* i, t, \mathbf{z}_t)$$

Layer 3 --- Dynamic structure

These create persistence and evolving conditions.

Examples:

$$X_t - \mu = \phi(X_{t-1} - \mu) + \varepsilon_t$$

and:

$$X_{i,t} = \mu_{i,t} + \beta^* i S_t + \varepsilon^* i, t$$

The credibility of BTYT comes from combining all three:

Hard accounting and relational consistency + heterogeneous stochastic behavior + coherent temporal dynamics.

68. Long-term objective of this document

When BTYT Part I is frozen, this document should be sufficiently precise that a technically trained reader can understand:

- what is random;
- what is deterministic;
- what is conditional on what;
- which processes share causal inputs;
- which realizations remain independent;
- how time enters the system;
- which identities must hold exactly;
- how the generators are statistically audited.

When Part II is completed, the same document should additionally explain:

- how the credit target is constructed;
- what information is available at prediction time;
- how features are defined;
- how models estimate risk;
- how discrimination and calibration are measured;
- how experiments are compared and tracked.

The final objective is not mathematical complexity for its own sake.

It is to make the synthetic bank mathematically legible, reproducible, and auditable.